INST447 - Data Sources and Manipulation
Midterm Exam

Due: Nov 23 (Sun.) 23:59pm.

What to submit: 1. a .pdf document containing your answers to all parts of the exam, including tables, visualizations, and written responses. **2. a .py or .ipynb file** which contain your code that can run without errors and clearly produce outputs that match your PDF report. It should be well-commented to facilitate understanding of your approach.

Academic Integrity: You may use AI tools (ChatGPT, Claude, Cursor, GitHub Copilot, etc.) and consult course materials, documentation, and Internet resources. However, you may not collaborate with other students, share code or answers, or seek assistance from any other individuals.

Late Submission: 0% penalty for the first 24 hours. 20% penalty for 24-48 hours. Submissions will not be accepted more than 48 hours after the deadline. Late submissions are not eligible for bonus points. Exceptions may be granted for documented emergencies at the instructor's discretion.

Introduction: The Challenge of Measuring Text Complexity

Imagine you're an educator designing curriculum for elementary students. You need to select reading materials that are challenging enough to promote learning, but not so difficult that students become frustrated. How do you measure whether a text is "just right" for 3rd graders versus 4th graders?

This is the problem that the CZI Learning Commons tackled in their Literacy Evaluators project. They assembled a team of experienced educators to read hundreds of texts and annotate them for complexity. The resulting dataset contains >1k educational texts evaluated for grades 3 and 4, with annotations for sentence/vocabulary complexity tier 2 and 3 words, and Flesch-Kincaid grade level.

Your task in this midterm is to conduct a guided **exploratory data analysis** of this dataset. You'll investigate how consistent the annotations are, what makes a text complex, whether automated measures like Flesch-Kincaid align with expert judgment, and whether modern LLMs can replicate human annotation decisions. This is not just a technical exercise—it's a window into how educational content is evaluated, how expert judgment works, and what challenges arise when trying to systematize subjective human assessments.

1. **(15 pts) Understanding What We're Working With**

   Before diving into complex analyses, we need to understand the dataset itself. What's in it? What's missing? What patterns emerge from a simple overview? In real exploratory data analysis, this foundation step is critical. You can't interpret findings if you don't know the structure of your data, and you can't trust conclusions if you haven't examined what might be missing or inconsistent.

   Your task: Produce three sets of information that help us understand this dataset:

   Basic Structure Load the dataset (`text_complexity.csv`) and examine its dimensions and contents. How many texts are in the dataset? How many variables describe each text? What do the column names tell us about what was measured? You should notice something interesting: there are more rows than unique texts (identified by `Clear ID`). Some texts appear multiple times. We'll explore why shortly, but for now, just document what you observe.

   Missing Data Patterns Not all data is complete. Educational content annotation is labor-intensive, and some fields may not apply to all texts. Calculate which columns have missing values (real `NaN` values, not the string "not scored"). Which columns are missing data for more than 20% of texts? Think about why certain fields might have high missingness?

Complexity Landscape What does the complexity distribution look like? For both "Sentence Score" and "Vocabulary Score", show the counts across all categories (including "not scored"). You can present this as a table or visualize it (pie chart or bar chart work well). This gives us a sense of what the typical text looks like in this dataset. Are most texts slightly complex or very complex? Do annotators frequently use the "exceedingly complex" category, or is it reserved for rare cases?

What to submit: (1) Dataset dimensions and observation about duplicates, (2) Missing data summary (which columns, what percentage), (3) Complexity distributions including "not scored" (table or visualization).

2. **(15 pts) Decoding the Annotation Process**

   Now that we understand what's in the dataset, let's understand how it was created. The documentation mentions that texts were annotated by multiple people and scores were "consolidated using the Dawid-Skene method"–a statistical technique for combining multiple annotators' judgments when they disagree.

   But how many annotators actually looked at each text? This isn't explicitly in the data, but we can decode it. Look at the "Sentence Score Rationale" column. Instead of a single explanation, you'll see entries like:

```
ss_annotator_1: The text uses simple sentences...
ss_annotator_2: Mostly straightforward structure...
ss_annotator_3: Some compound sentences appear...
```

   Each annotator left their reasoning, prefixed with their identifier. This is our clue—we can count how many annotators contributed to each text's evaluation.

   Your task: Extract the annotator count from the rationale text. You'll need to use **regular expressions** to find patterns like `ss_annotator_1:`, `ss_annotator_2:`, `vocab_annotator_1:`, etc. For each text, count how many unique annotators contributed to the Sentence Score and how many contributed to the Vocabulary Score. Once you've extracted these counts, show us the distribution. How many texts had 2 annotators? 3? 4? More? Is the pattern the same for Sentence complexity and Vocabulary complexity annotations? Create a visualization (bar chart is fine) showing the annotator count distribution for Sentence and Vocabulary annotations.

   Note: You could also use LLM to analyze whether annotators agreed or disagreed with each other by examining their rationale text, but that's out of scope for this exam. Feel free to explore it in your free time if interested!

   What to submit: (1) Distribution of annotator counts (how many texts have 2, 3, 4+ annotators), (2) Visualization showing the distribution, (3) Brief observation about the pattern.

3. **(20 pts) Is Tier Classification Consistent?**

   With an understanding of the annotation process, we can now ask: Do annotators agree on how to classify vocabulary? The dataset distinguishes between Tier 2 words (general academic vocabulary like "analyze" or "contribute") and Tier 3 words (domain-specific terms like "photosynthesis" or "denominator"). But is this distinction clear-cut? Or do different annotators sometimes disagree about whether a word belongs in Tier 2 versus Tier 3?

   Your task: Investigate annotator consistency in two ways.

   A. **(10 pts) Consensus Within Grade Level**
      Focus on Grade 3 texts. For the same grade level, do we see the same word classified differently across texts? Parse the "Tier 2 Words" and "Tier 3 Words" columns (they're comma-separated lists), and explode them so each word gets its own row. Now find words that appear in **both** Tier 2 and Tier 3 across different Grade 3 texts. Report how many words show this inconsistency. Give us a few examples showing the pattern you found.

B. **(10 pts) Grade-Dependent Classification**

Now compare across grades. Are there words that annotators classify as Tier 3 when evaluating for Grade 3, but Tier 2 when evaluating for Grade 4 (or vice versa)? This would suggest that "tier" isn't just a property of the word itself, but depends on the grade context—what's domain-specific for 3rd graders might be general academic vocabulary for 4th graders. Find and report examples of words that switch tiers between Grade 3 and Grade 4 annotations.

What to submit: (1) Analysis A: Count and examples of words appearing in both tiers within Grade 3, (2) Analysis B: Count and examples of words switching tiers between grades, (3) Brief interpretation: If you find substantial tier-switching in either analysis, what does this tell us about vocabulary classification? Is it objective or subjective? This is important context for everything else we'll analyze—these annotations represent expert human judgment, but that doesn't mean they're perfectly consistent.

4. **(15 pts) What Predicts Vocabulary Complexity?**

Now that we've looked at individual columns, let's examine relationships between columns. In particular, let's test whether the two measures of vocabulary (complexity score vs. Tier 2/3 words) are related. Specifically: Are Tier 3 words a good predictor of vocabulary complexity? The hypothesis is straightforward—texts with more domain-specific technical terminology should be rated as more complex. But does the data support this?

This is a common pattern in exploratory analysis: **we come up with hypotheses about what drives an outcome, and we test it empirically**.

Your task: Investigate the relationship between Tier 3 word count and Vocabulary Score. First, count how many Tier 3 words appear in each text (parse the comma-separated list). Then group texts by their Vocabulary Score and calculate the mean Tier 3 word count for each complexity level. Finally, visualize this relationship. A bar chart with complexity categories on the x-axis and mean Tier 3 count on the y-axis works well. You could also use box plots to show the distribution.

What to look for: Is there a clear progression? Do "slightly complex" texts have fewer Tier 3 words than "very complex" texts? Or is the relationship messy and unclear? If Tier 3 words are strongly predictive, it validates that despite the annotation inconsistencies we found in Part 3, the overall pattern makes sense. If there's no relationship, it raises questions about how complexity scores were assigned.

What to submit: (1) Grouped means showing Tier 3 word counts by complexity level, (2) Visualization, (3) Brief interpretation of the pattern.

5. **(15 pts) Can LLM Annotate Vocabulary Tiers?**

Now for a modern twist: Could we use a Large Language Model to perform this annotation task? Given all the work involved in having expert educators read texts and identify Tier 2 and Tier 3 words, it's natural to ask whether an LLM could do the same task.

Your task: Test whether an LLM can identify Tier 2 and Tier 3 words in a text the way human annotators do. Select 3–5 texts that have both Tier 2 and Tier 3 words annotated. For each text, provide the LLM with the text and grade level, then ask it to identify Tier 2 and Tier 3 words.

You should use the tier definitions from the Learning Commons documentation:

Structure your prompt to explain these definitions, provide the grade level and text, and request two lists of words. Once you have LLM annotations for your test texts, compare them with human annotations: How many words did both LLM and humans identify? How many did LLM miss that humans caught? How many did LLM add that humans didn't include? Did LLM sometimes classify a word as Tier 2 when humans said Tier 3 (or vice versa)?

You may want to explore different prompt phrasings to see what works best. You can use any LLM you like (GPT, Claude, Qwen, etc.), but be sure to document which one you used and the exact prompt text.

What we're looking for: Not necessarily high agreement—in fact, based on Part 3, we know human annotators don't always agree either. What's interesting is *how* the LLM's judgments differ. Does it tend to identify fewer words? More words? Does it systematically misclassify certain types of terms?

What to submit: (1) Your prompt (the exact text you sent to the LLM), (2) For each of your 3–5 test texts: (a) human annotation, (b) LLM annotation, (c) overlap analysis, (3) Brief interpretation of how LLM performance compares to humans.

6. **(10 pts) A Sanity Check on Cross-Grade Ratings**

   Let's step back and check whether the ratings make logical sense. You may have noticed that many texts appear twice in the dataset—once evaluated for Grade 3 and once for Grade 4 (same `Clear ID`). This design lets us ask: **How do annotators expect the same text to be perceived by students at different grade levels?**

   Your task: Investigate the pattern of complexity ratings across grades. Filter to texts that appear in both grades. Pivot them so you have one row per text with both complexity ratings. Convert the complexity scores to numbers so you can compare them (slightly=1, moderately=2, very=3, exceedingly=4, not scored=-1). Now count different patterns and report what you find. Examine some examples of unexpected patterns—what do you notice about them?

   What to submit: (1) Pattern analysis with counts, (2) Examples of any unexpected cases, (3) Your interpretation of whether the ratings make logical sense.

7. **(10 pts) AI Assistant Reflection**

   If you used AI assistants (ChatGPT, Claude, Cursor, Copilot, etc.): Which parts did you use AI for? What prompts or questions did you ask the AI? How did you verify the correctness of AI-generated code? What did you learn from using AI for this exam? Did the AI give any incorrect or confusing answers? How did you handle that?

   If you did NOT use AI: What was your approach to solving the problems? What resources did you use (documentation, online tutorial)? Why did you choose not to use AI?

   Please structure your answer as 2–3 paragraphs with examples. *This is the **ONLY** question where you should **NOT** use AI to write your response.*

8. **(BONUS up to 10 pts) The Flesch-Kincaid Mystery**

   The Flesch-Kincaid formula predicts grade level based on sentence length and syllable count. For a text appropriate for 3rd graders, we'd expect FK ≈ 3. But when we calculated the mean FK score in this dataset, we got 7.2–suggesting the texts are appropriate for 7th graders, not 3rd/4th graders! Why does FK miss the mark so badly? Is the formula flawed? Are the texts mislabeled? Does FK ignore important factors like vocabulary complexity?

   You can answer with just a written hypothesis, or you can dig into the data to test your hypothesis with code. We're interested in your reasoning and how you support it.

Tips for Success

Start early Some questions require OpenAI API calls which can be slow.

Document as you go Take notes while coding.

Test your code Make sure it runs from top to bottom without errors.

Think critically Exploratory Data Analysis is about interpretation, not just computation.

Good luck! This is exploratory data analysis—there's rarely one "right" answer. We're evaluating your ability to investigate data systematically, interpret findings thoughtfully, and communicate clearly.